SIT720 - Machine Learning

6.1 PASS TASK

Linear Regression and Logistic Regression

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Dataset: UCI Automobile Dataset
Random seed: 42

Report draft for review and adaptation before submission

Part A - Lesson Review and Reflection

A1. Key Learning from Week 5: Regression and Prediction Models

Two concepts that I found especially important are regression analysis/model fitting and the separation of training and testing data. Regression analysis models how a continuous dependent variable changes as one or more independent variables change. During fitting, the algorithm estimates parameters that minimise a loss, such as squared error. This matters because a fitted model turns observed data into a quantitative relationship that can be used for prediction. For example, a vehicle-price model can learn how engine size, curb weight and horsepower relate to price and then estimate the price of a new vehicle.

The second concept is evaluating a model on data that was not used for fitting. Training performance shows how well the model explains known examples, but test performance better estimates how it will behave on unseen cases. Metrics such as MAE, RMSE and R-squared describe different aspects of regression error. This separation is important because a model can memorise patterns or noise and appear excellent on training data while performing poorly in practice. In the Automobile analysis, comparing training and test scores helped identify overfitting in the full Linear Regression model.

A2. Key Learning from Week 6: Supervised Learning and Model Optimisation

Logistic Regression and regularisation are two important Week 6 concepts. Logistic Regression is a supervised classification method that models the log-odds of a class as a linear combination of predictors and converts the result to a probability using the logistic function. It is useful when the target is categorical rather than continuous. For example, the Automobile data can be converted into a binary task where a car is labelled luxury if its price is above the dataset median. The predicted probability can then be converted to a class using a threshold such as 0.5.

Regularisation controls model complexity by adding a penalty to the optimisation objective. L1 regularisation penalises the absolute size of coefficients and can set some coefficients to zero, which can also perform feature selection. L2 regularisation penalises squared coefficients and tends to shrink all coefficients smoothly. Regularisation is important when many correlated or one-hot encoded predictors are available because an unrestricted model can fit noise and produce unstable coefficients. In this task, Ridge and Lasso improved the test performance of price prediction, while L1/L2 Logistic Regression reduced the extreme behaviour of the unregularised classifier.

A3. Applying Regression and Classification to Data Analytics

Regression and classification answer different real-world questions from the same dataset. A regression model is suitable when the outcome is continuous. In the Automobile dataset, Linear Regression can estimate a numerical vehicle price from attributes such as curb weight, engine size, width and horsepower. Classification is suitable when the outcome is a category. After creating LuxuryCar from the median price, Logistic Regression can estimate the probability that a vehicle belongs to the luxury class and then assign a binary label.

Model complexity must be controlled in both settings. Adding many variables can reduce training error, but it can also increase variance, especially when the dataset is small and categorical variables expand into many dummy columns. A model that is too simple underfits because it cannot represent useful relationships; a model that is too complex overfits because it learns sample-specific noise. Reliable machine learning therefore requires a balance between fit and generalisation. Feature selection and regularisation are practical ways to control this balance. In this analysis, reducing the regression feature space and using L1/L2 penalties produced stronger test performance than relying only on the unregularised full model.

Part B - Problem Solving Practical Task

1. Data Preparation and Encoding

The UCI Automobile dataset contains 205 records and 26 columns (25 predictors plus price in the working table). Missing values were present in normalized_losses (41), bore (4), stroke (4), price (4), num_of_doors (2),

horsepower (2) and peak_rpm (2). Because price is the supervised target, the four records with unknown price were removed rather than imputed, leaving 201 usable vehicles. The median price was $10,295; LuxuryCar was defined as 1 for prices above this value and 0 otherwise.

The predictors comprised 15 numeric and 10 categorical variables. Numeric missing values were median-imputed and standardised with StandardScaler. Categorical missing values were replaced by the most frequent category and then one-hot encoded with one reference level removed. Preprocessing was placed inside scikit-learn pipelines so imputation, scaling and encoding were learned only from training data. A stratified 80/20 split produced 160 training and 41 test records; the test set contained 21 non-luxury and 20 luxury vehicles.

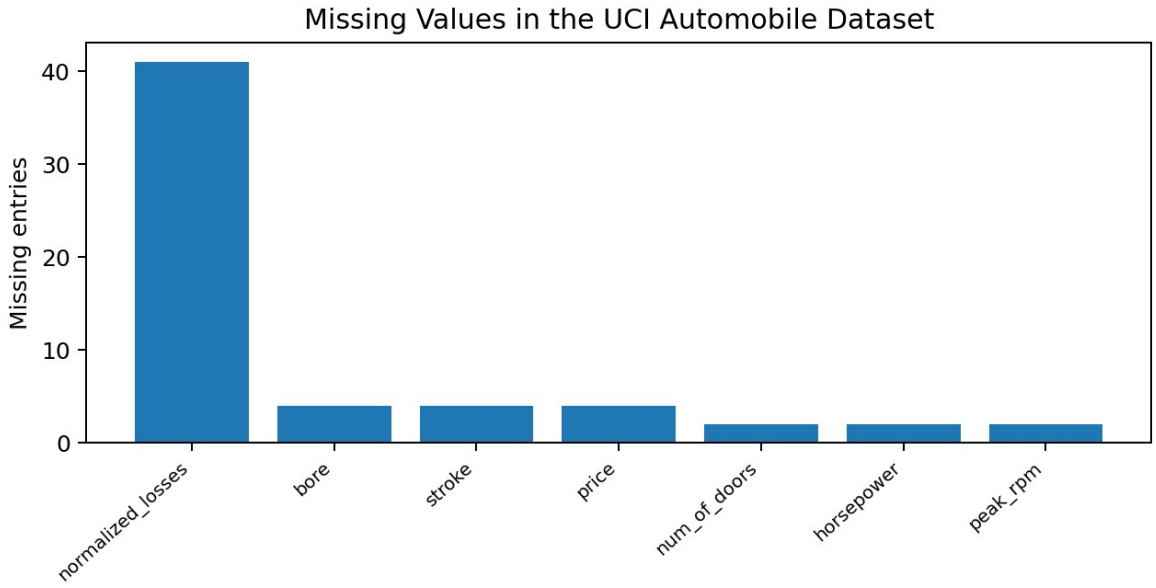


Figure 1. Missing values before preprocessing.

```
numeric_pipe = Pipeline([('imputer', SimpleImputer(strategy='median')),
                         ('scaler', StandardScaler())])
categorical_pipe = Pipeline([('imputer', SimpleImputer(strategy='most_frequent')),
                             ('onehot', OneHotEncoder(drop='first', handle_unknown='ignore'))])
```

2. Linear Regression: Vehicle Price Prediction

A Linear Regression model was trained on all preprocessed predictors. The fitted intercept was $24,625.75. Because numeric features were standardised and categorical variables were dummy-coded, this intercept represents the reference-category vehicle at the mean of the numeric training variables; it should not be interpreted as the price of a typical raw vehicle.

Numeric feature	Coefficient ($ per 1 SD)	Direction
curb_weight	3,515	Positive
engine_size	2,512	Positive
length	-2,049	Negative
width	1,741	Positive
horsepower	1,494	Positive
compression_ratio	-1,463	Negative

The largest standardised numeric effects were curb weight (+$3,515 per standard deviation), engine size (+$2,512), length (-$2,049), width (+$1,741), horsepower (+$1,494) and compression ratio (-$1,463), holding other predictors constant. The positive effects of weight, engine size, width and horsepower are reasonable because larger and more powerful vehicles are often more expensive. The negative conditional coefficient for length should not be read causally: length, wheelbase, width, curb weight and engine size are correlated, so a coefficient represents the effect after the other correlated measurements are held fixed. Some make/category

dummy variables also had large coefficients because they are measured relative to a reference category and some categories contain few vehicles.

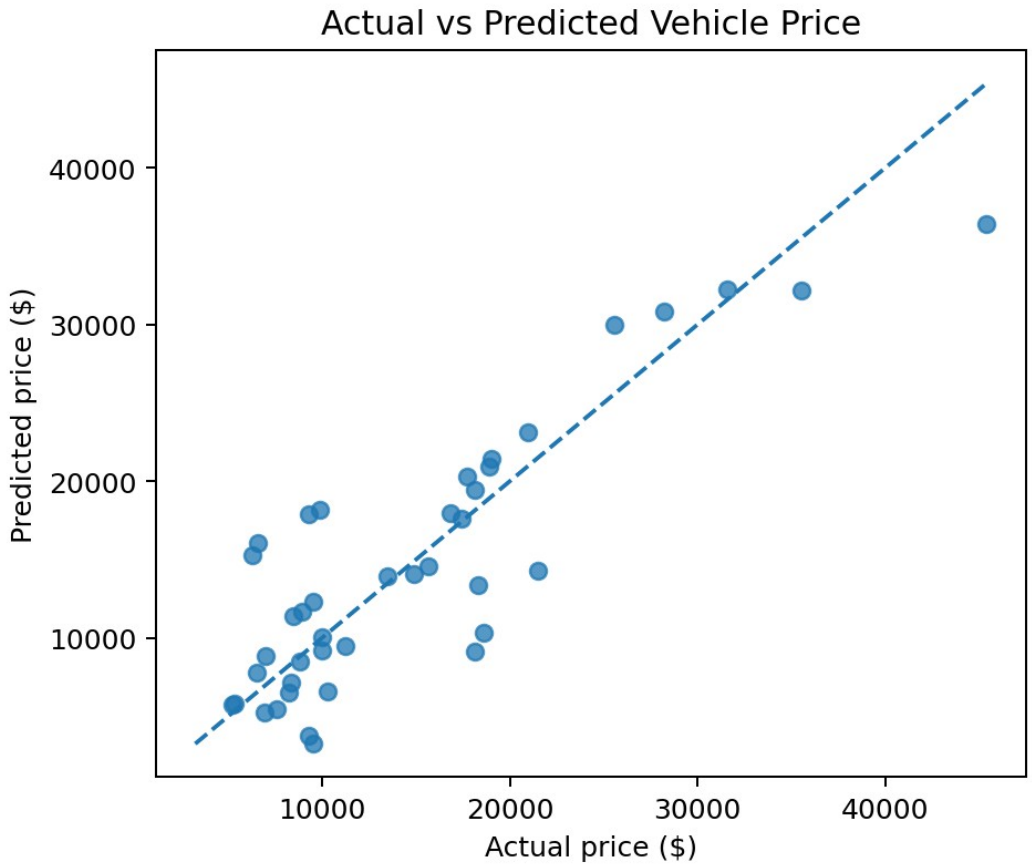


Figure 2. Actual and predicted prices on the 41-vehicle test set.

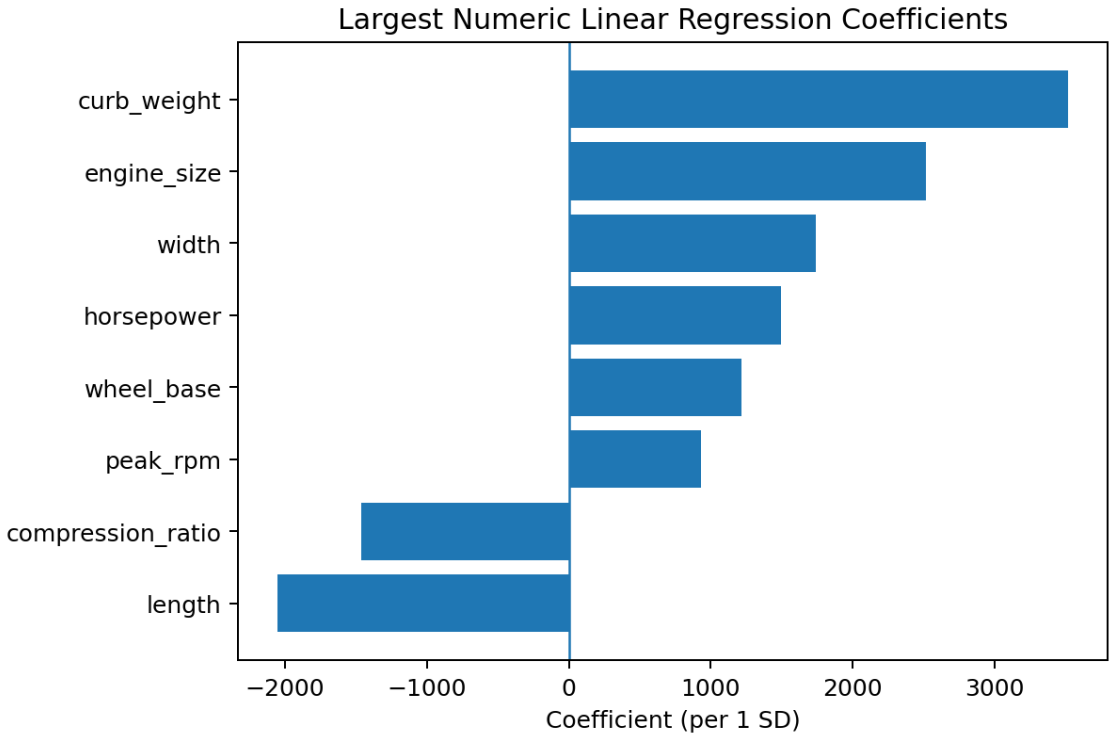


Figure 3. Largest standardised numeric coefficients in the full Linear Regression model.

3. Evaluating Linear Regression Performance

Set	MAE ($)	MSE ($²)	RMSE ($)	R^2
Train	802	1,314,390	1,146	0.978
Test	3,335	19,861,470	4,457	0.738

MAE gives the average absolute dollar error and is easy to interpret. MSE squares each error, so large mistakes receive a much larger penalty. RMSE is the square root of MSE and expresses the error in dollars, while R-squared is the proportion of target variance explained relative to a mean-only baseline. On the test set, the model achieved MAE = $3,335, RMSE = $4,457 and R-squared = 0.738. Therefore, it explains a substantial proportion of price variation, but prediction errors are still material.

The training R-squared of 0.978 is much higher than the test R-squared of 0.738, and training RMSE ($1,146) is far below test RMSE ($4,457). This is evidence of overfitting rather than underfitting. Likely causes are the small sample size, many one-hot encoded categories, rare makes, correlated vehicle dimensions, possible outliers and nonlinear price relationships that a single linear function cannot represent.

4. Logistic Regression: Luxury Car Classification

Logistic Regression is appropriate because LuxuryCar is binary and the model directly estimates P(LuxuryCar = 1). Unlike Linear Regression, which predicts an unrestricted continuous value, Logistic Regression models log-odds and uses a logistic transformation so probabilities lie between 0 and 1. A 0.5 threshold was used for class labels. The same preprocessed input features and train/test records were used as in the price model.

The most influential coefficients in the unregularised classifier included highway_mpg (-62.7), curb_weight (+57.5), width (+44.1), compression_ratio (+41.9), city_mpg (-41.5) and the mpfi fuel-system indicator (+38.0). Larger curb weight and width increase the estimated luxury odds, while higher fuel economy decreases them. The very large coefficient magnitudes and probabilities close to 0 or 1 indicate near-separation and overconfidence in this small dataset, which is one reason to evaluate regularised models in Question 7.

```
logistic_model = Pipeline([
    ('preprocess', preprocessor),
    ('model', LogisticRegression(penalty=None, max_iter=5000))
])
proba = logistic_model.predict_proba(X_test)[:, 1]
labels = (proba >= 0.5).astype(int)
```

5. Evaluating Logistic Regression Performance

Accuracy	Precision	Recall	F1-score	ROC-AUC
0.902	0.833	1.000	0.909	1.000

Accuracy measures the overall fraction of correct classifications. Precision measures how many predicted luxury cars are truly luxury. Recall measures how many true luxury cars are found, and F1 combines precision and recall using their harmonic mean. ROC-AUC evaluates ranking quality across all possible thresholds. The test accuracy was 0.902, precision 0.833, recall 1.000 and F1 0.909. The ROC-AUC was 1.000 on this particular 41-record split, but this should be interpreted cautiously because the test sample is small and the unregularised probabilities are extremely confident.

The confusion matrix contained 17 true negatives, 4 false positives, 0 false negatives and 20 true positives. The model therefore identified every luxury car, but four non-luxury vehicles were incorrectly promoted to luxury. Its non-luxury specificity was 17/21 = 0.810, so performance was not equally balanced across both classes: recall for the luxury class was stronger than recognition of the non-luxury class.

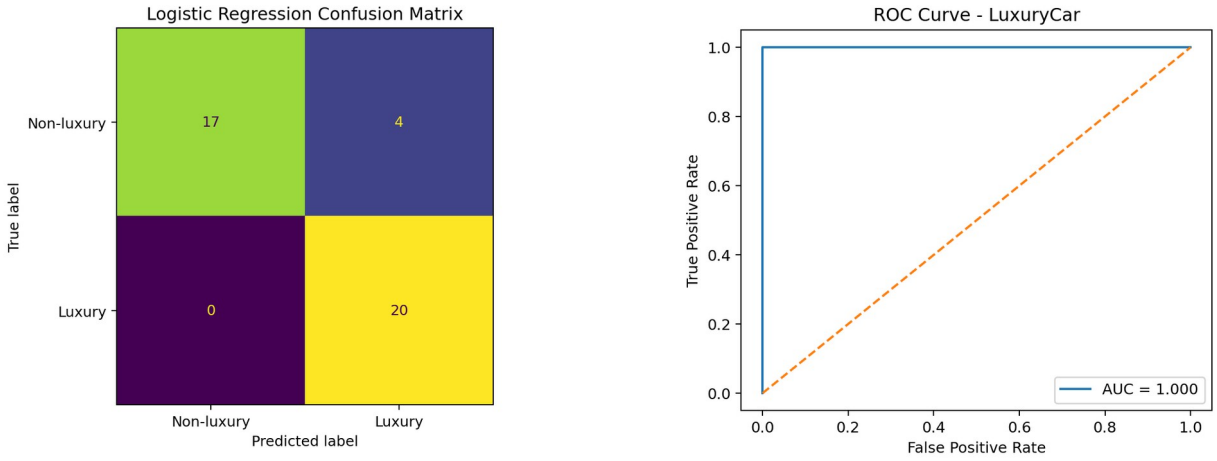


Figure 4. Confusion matrix (left) and ROC curve (right) for the unregularised Logistic Regression model.

6. Feature Selection and Model Generalisation

Model	Train score	Test score	Interpretation
Linear - all features	R^2 0.978	R^2 0.738	Large generalisation gap
Linear - selected 15	R^2 0.829	R^2 0.903	Best regression generalisation
Logistic - all features	Acc 0.994	Acc 0.902	Very high training fit
Logistic - selected 15	Acc 0.963	Acc 0.902	Same test accuracy, lower train fit

The selected feature models used SelectKBest with k = 15, fitted inside the training pipeline. For regression, feature selection reduced training R-squared from 0.978 to 0.829 but increased test R-squared from 0.738 to 0.903 and reduced test RMSE to $2,712. This is strong evidence that additional features did not automatically improve the model; the full model had learned high-variance relationships that did not transfer well to unseen vehicles.

For classification, selecting 15 features reduced training accuracy from 0.994 to 0.963 while test accuracy remained 0.902; test AUC decreased from 1.000 to 0.974. The selected model is less fitted to the training sample but does not improve thresholded test accuracy. Underfitting occurs when a model is too simple and performs poorly on both training and test data. Overfitting occurs when training performance is much stronger than test performance. Here, the full regression model shows clear overfitting, while the selected regression model generalises more effectively.

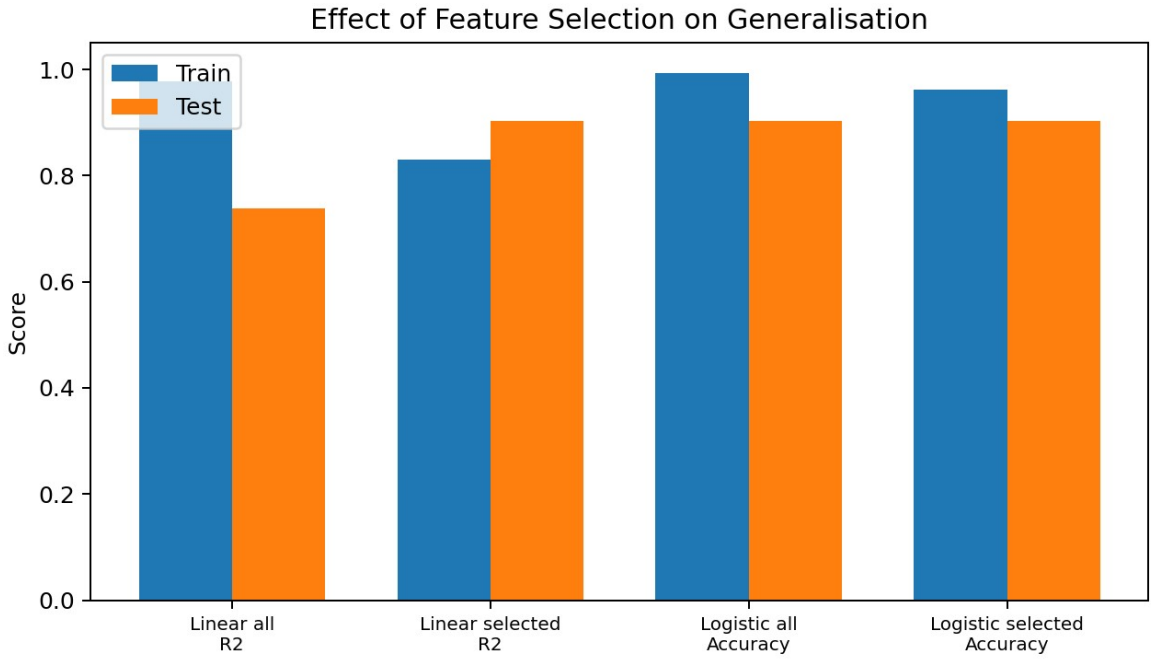


Figure 5. Training and test scores before and after selecting 15 transformed features.

7. Regularisation for Improving Model Performance

Regression model	Best hyperparameter	Test RMSE ($)	Test R^2
Linear (none)	-	4,457	0.738
Ridge (L2)	alpha = 0.1	3,322	0.855
Lasso (L1)	alpha = 10	2,855	0.893

Logistic model	Best C	Test accuracy	Precision	Recall	F1	AUC
No penalty	-	0.902	0.833	1.000	0.909	1.000
L1	1	0.927	0.905	0.950	0.927	0.986
L2	1	0.927	0.905	0.950	0.927	0.990

Ridge uses an L2 penalty and shrinks coefficients toward zero, while Lasso uses an L1 penalty and can set weak coefficients exactly to zero. Five-fold cross-validation selected alpha = 0.1 for Ridge and alpha = 10 for Lasso. Both regularised regression models improved test performance relative to ordinary Linear Regression. Lasso achieved the strongest regularised result (R-squared 0.893, RMSE $2,855), indicating that suppressing weak/redundant coefficients reduced variance substantially.

For Logistic Regression, five-fold cross-validation selected C = 1 for both L1 and L2 penalties. Both increased test accuracy from 0.902 to 0.927 and changed the confusion matrix to 19 true negatives, 2 false positives, 1 false negative and 19 true positives. L2 achieved AUC 0.990 and L1 achieved 0.986. Although the unregularised model had a perfect test AUC on this small split, its lower accuracy and near-binary probabilities indicate an unstable, overconfident fit. The regularised classifiers provide a more balanced decision boundary and smaller train-test gap.

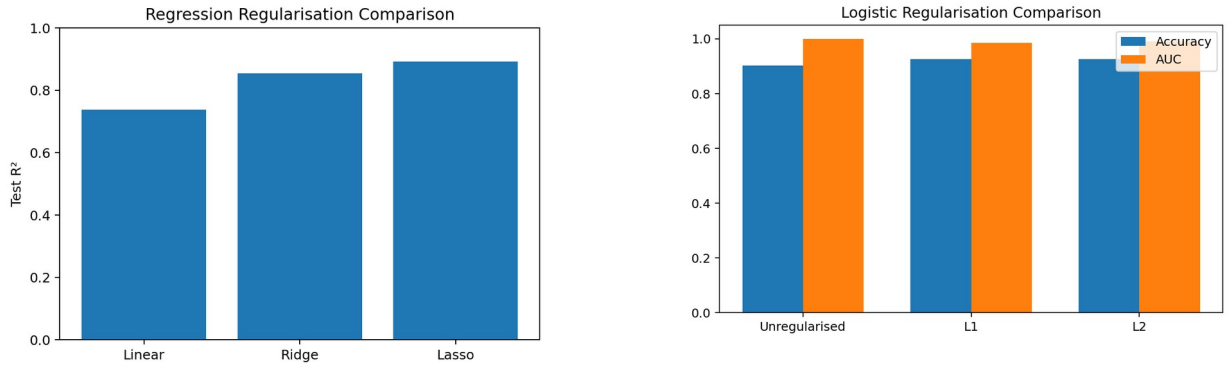


Figure 6. Effect of regularisation on regression (left) and classification (right).

Conclusion

The Automobile dataset demonstrates why evaluation on unseen data is essential. Ordinary Linear Regression achieved a high training fit but showed clear overfitting; feature selection and regularisation produced much stronger test behaviour. For binary LuxuryCar prediction, the unregularised Logistic Regression model achieved high recall but was overconfident and generated four false positives. L1 and L2 regularisation improved test accuracy to 92.7% and produced a more balanced confusion matrix. Overall, the results support using preprocessing pipelines, train-only feature selection, cross-validation and regularisation to improve model generalisation rather than relying on training performance alone.

GenAI Acknowledgement - edit to match your actual use before submission

ChatGPT was used to assist with planning the analysis, generating initial code scaffolding, and drafting/editing explanatory text. The generated material must be critically reviewed, understood, tested and modified by the student before submission. The final acknowledgement should be updated so it accurately describes how GenAI was actually used, consistent with the assessment instructions.

References

[1] J. Schlimmer, “Automobile,” UCI Machine Learning Repository, 1985, doi: 10.24432/C5B01C.

[2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.